

Wigner–Ville Distribution with Multiple Quaternion Correlations via the Offset Linear Canonical Transform

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Abstract—Statistical methods for quaternion enhancement, which are based on multiple quaternion correlations, have found extensive applications in areas such as unmanned aerial vehicle positioning and multiplicative noise removal. However, research on the Wigner–Ville distribution associated with the quaternionic offset linear canonical transform (WVD-QOLCT) fields only utilizes one quaternion correlation based on the conjugate of all imaginary parts of quaternions. In this study, to fully exploit the multiple correlations inherent in quaternions, we propose a novel Wigner–Ville distribution with multiple quaternion correlations via the offset linear canonical transform (WVD-QCOLCT), which introduces the multiple correlation characteristics of quaternions into the WVD-QOLCT. Subsequently, we derive some properties of the WVD-QCOLCT, such as the inverse, the relationship between multiple quaternion correlations in the WVD-QCOLCT, and the uncertainty principle. Finally, a detection algorithm for quaternion signals is proposed to evaluate the performance of the WVD-QCOLCT.

Index Terms—multiple quaternion correlations, wigner-ville distribution, offset linear canonical transform, signal detection

I. INTRODUCTION

The traditional Fourier transform (FT) can convert signals from the time domain to the frequency domain, thereby exposing the frequency composition of the signal. However, non-stationary signals have fixed time-frequency resolutions that cannot be adapted to changes in signal frequency over time, resulting in limitations in the analysis results. To overcome this problem, time-frequency analysis methods have emerged, attempting to simultaneously display signal characteristics across both the time and frequency domains, where the Wigner–Ville distribution (WVD) stands as a classic example [1]. The linear canonical transform (LCT) [2] integrates diverse transformation forms, such as the FT and fractional Fourier transform (FRFT), and transforms signals through specific symplectic matrices. Building on this foundation, the offset linear canonical transform (OLCT) incorporates the offset parameters. These parameters enable a flexible adjustment

of the signal distribution across the time-frequency plane, allowing for signal stretching, compression, and offset to varying extents, while also offering additional degrees of freedom for signal analysis. [3]. This flexibility makes OLCT suitable for broad application prospects in time–frequency offset detection [4], signal filtering [5], and multichannel sampling with limited signals [6]. However, its deep integration with other advanced signal processing theories requires further exploration.

Quaternions, a high-dimensional extension of complex numbers, were proposed by Hamilton in 1843 [7]. According to multiple conjugate forms of quaternions, reference [8] proposes using covariance and covariance matrices to obtain complete second-order statistical information. Reference [9] focuses on the enhanced statistics of quaternion random variables, highlighting that quaternion statistics based on multiple conjugates and combinations are the key to quaternion learning machines in two application examples: unmanned aerial vehicle 3D direction tracking and 3D direction variable denoising.

The LCT has demonstrated significant effectiveness in analyzing WVD [10] and ambiguity functions (AF), proving particularly useful for handling non-stationary signal processing applications [11]. In [12], quaternions were introduced into the WVD-LCT, and a two-dimensional quaternion Wigner distribution (QLWD) was proposed and applied for signal detection. The Parametric quaternion Wigner–Ville distribution (PQWD) was proposed in [13], which improves the flexibility of the WVD-QLCT. By integrating the WVD, QFT, and OLCT, El Kassimi et al. took the lead in putting forward the WVD based on the QOLCT (WVD-QOLCT) in 2019 [14]. Reference [15] defines the convolution and related operators of the QOLCT and provides the convolution and related theorems of the WVD-QOLCT. Reference [16] developed the logarithmic, Hardy, and Beurling uncertainty principles for the WVD-QOLCT, which enriched the theoretical system of uncertainty. These achievements provide new tools and theoretical foundations for the analysis of non-stationary signals. However, existing research is mainly limited to a single quaternion conjugate form and has not fully utilized

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the rich representation capabilities provided by the multiple correlations of quaternions.

This study aims to extend the WVD-QOLCT by incorporating multiple quaternion correlations, thereby proposing a new transformation called the WVD-QCOLCT. In Section II, this paper reviews the key properties of quaternion algebra and introduces the WVD-QOLCT. In Section III, the WVD-QCOLCT is established, the relationships of five quaternion correlations are derived, and some important properties of the WVD-QCOLCT are derived. In Section IV, the application of the WVD-QCOLCT for signal detection is presented. Section V provides a summary of this study.

II. PRELIMINARIES

A. Quaternion algebra

As preliminary knowledge of the paper, this section offers the following fundamental knowledge about quaternion algebra. Unlike complex numbers, which contain only one imaginary part, quaternions introduce three mutually orthogonal imaginary units i, j, k thus expanding the number field from two dimensions to four dimensions. The standard form of a quaternion can be expressed as

$$q = q_0 + iq_i + jq_j + kq_k \in \mathbb{H}, (q_n \in \mathbb{R}, n = 0, i, j, k). \quad (1)$$

Here, the elements i, j, k adhere to Hamilton's multiplication rules:

$$\begin{aligned} i^2 = j^2 = k^2 = ijk = -1, \quad ij = -ji = k, \\ jk = -kj = i, \quad ki = -ik = j. \end{aligned}$$

This set of rules implies that quaternion multiplication fails to satisfy the commutative law, that is, $pq \neq qp$ for $p, q \in \mathbb{H}$. Its essence stems from the noncommutativity of rotation operations in three-dimensional space, which also establishes the core position of quaternions in geometry and physics. Using only a single conjugate cannot represent all the imaginary components of the quaternions. Thus, for a quaternion $q = q_0 + iq_i + jq_j + kq_k$, reference [17] defines its multiple conjugates $q^{\eta*}$ ($\eta = 0, 1, i, j, k$), where

$$\begin{aligned} q^{0*} &= q_0 + iq_i + jq_j + kq_k, & q^{1*} &= q_0 - iq_i - jq_j - kq_k, \\ q^{i*} &= q_0 - iq_i + jq_j + kq_k, & q^{j*} &= q_0 + iq_i - jq_j + kq_k, \\ q^{k*} &= q_0 + iq_i + jq_j - kq_k. \end{aligned}$$

According to quaternion multiple conjugates, we have

$$\begin{aligned} (pq)^{\eta*} &= q^{\eta*} p^{\eta*}, (pq)^{0*} = pq, \\ (p+q)^{\eta*} &= p^{\eta*} + q^{\eta*}, (p^{\eta*})^{\eta*} = p, (\eta = 1, i, j, k) \end{aligned}$$

and

$$\begin{aligned} q_0 &= \frac{1}{2} (q^{0*} + q^{1*}), & q_i &= \frac{1}{2} (q^{0*} - q^{1*}), \\ q_j &= \frac{1}{2} (q^{0*} - q^{j*}), & q_k &= \frac{1}{2} (q^{0*} - q^{k*}). \end{aligned}$$

Using the conjugate q^{1*} of a quaternion q , we can obtain the definition of the norm of a quaternion

$$|q|_Q = \sqrt{qq^{1*}} = \sqrt{q_0^2 + q_i^2 + q_j^2 + q_k^2}.$$

For a quaternion $q \neq 0$, we can define its inverse $q^{-1} = \frac{q^{1*}}{|q|_Q^2}$. For any quaternion q whose imaginary part is non-zero (i.e. q_i, q_j, q_k are not all zero), it can be further expressed in exponential form by decomposing it into real and pure imaginary parts $q = |q|e^{v\theta}$, where $|q|$ denotes the norm of q , $v = \frac{q_i i + q_j j + q_k k}{\sqrt{q_i^2 + q_j^2 + q_k^2}}$, and $\theta = \arccos\left(\frac{q_0}{|q|}\right)$. $e^{v\theta}$ can be expanded into

$$e^{v\theta} = \cos(\theta) + v \sin(\theta).$$

A quaternion function is defined as $f(t) : \mathbb{R}^2 \rightarrow \mathbb{H}$. The value of a quaternion function can be expressed as

$$f = f_0 + if_i + jf_j + kf_k,$$

where $f_n(t) : \mathbb{R}^2 \rightarrow \mathbb{H}$, for $n = 0, i, j, k$. The following is the definition of the inner product of the quaternion functions:

$$\langle f, g \rangle = \int_{\mathbb{R}^2} f(t) g^{1*}(t) dt.$$

The following serves as the definition of the norm of quaternion functions:

$$|f|_{2,Q} = \left(\int_{\mathbb{R}^2} |f|_Q^2 dt \right)^{1/2}.$$

B. WVD-QOLCT

In signal processing, different QOLCT parameters can be flexibly selected according to the specific needs of the limit, and signals can be analyzed and processed from a broader perspective. The WVD-QOLCT stands as a crucial method for handling non-stationary signals within the transform domain.

Definition 1. (QOLCT [18]).

Let $A_l = \begin{bmatrix} a_l & b_l \\ c_l & d_l \end{bmatrix} \begin{matrix} \tau_l \\ \omega_l \end{matrix}$, with $a_l, b_l, c_l, d_l, \tau_l, \omega_l \in \mathbb{R}$ such that $a_l d_l - b_l c_l = 1$, for $l = 1, 2$. This study only considers the case of $b_1, b_2 \neq 0$. The QOLCT is defined as

$$\begin{aligned} \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{f(t)\}(u) &= \int_{\mathbb{R}^2} K_{A_1}^{\lambda}(t_1, u_1) f(t) K_{A_2}^{\mu}(t_2, u_2) dt, \\ b_1, b_2 &\neq 0, \end{aligned} \quad (2)$$

where λ, μ belong to the set of purely imaginary quaternions.

$$\begin{aligned} K_{A_1}^{\lambda}(t_1, u_1) &= \frac{1}{\sqrt{\lambda 2\pi b_1}} \times \\ &e^{\lambda(a_1 t_1^2 - 2t_1(u_1 - \tau_1) - 2u_1(d_1 \tau_1 - b_1 \omega_1) + d_1(u_1^2 + \tau_1^2)) \frac{1}{2b_1}}, b_1 \neq 0, \end{aligned} \quad (3)$$

and

$$\begin{aligned} K_{A_2}^{\mu}(t_2, u_2) &= \frac{1}{\sqrt{\mu 2\pi b_2}} \times \\ &e^{\mu(a_2 t_2^2 - 2t_2(u_2 - \tau_2) - 2u_2(d_2 \tau_2 - b_2 \omega_2) + d_2(u_2^2 + \tau_2^2)) \frac{1}{2b_2}}, b_2 \neq 0. \end{aligned} \quad (4)$$

Definition 2. (WVD-QOLCT [14]).

Let $A_l = \begin{bmatrix} a_l & b_l \\ c_l & d_l \end{bmatrix} \begin{matrix} \tau_l \\ \omega_l \end{matrix}$ such that $a_l d_l - b_l c_l = 1$, for $l = 1, 2$. This paper only considers the case of $b_1, b_2 \neq 0$.

The WVD associated with the WVD-QOLCT of a quaternion signal f is defined

$$\begin{aligned} \mathcal{W}_{f,g}^{A_1,A_2}(t,u) = \\ \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \bar{g}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2,u_2) ds, \\ b_1, b_2 \neq 0, \end{aligned} \quad (5)$$

where $\eta = 0, 1, i, j, k$, $K_{A_1}^\lambda(s_1, u_1)$, and $K_{A_2}^\mu(s_2, u_2)$ are given by (3) and (4), respectively.

III. WVD-QCOLCT

This section provides a definition of WVD-QCOLCT. Furthermore, the relationships between multiple types of WVD-QCOLCTs are presented. Then, some basic properties of the WVD-QCOLCT are derived. Finally, the uncertainty principle of the WVD-QCOLCT is presented.

A. The definition of the WVD-QCOLCT

Using only the correlation $qp^*(q, p \in \mathbb{H})$ of quaternions is insufficient to fully characterize the second-order statistical information of quaternions. Therefore, we use multiple quaternion correlations $qp^{\eta*}(q, p \in \mathbb{H})$ to define the WVD-QCOLCT.

Definition 3. (WVD-QCOLCT).

Let $A_l = \begin{bmatrix} a_l & b_l \\ c_l & d_l \end{bmatrix} \begin{bmatrix} \tau_l \\ \omega_l \end{bmatrix}$, with $a_l, b_l, c_l, d_l, \tau_l, \omega_l \in \mathbb{R}$ such that $a_l d_l - b_l c_l = 1$, for $l = 1, 2$. We only consider the case of $b_1, b_2 \neq 0$. The WVD-QCOLCT of a quaternion signal f is given by

$$\begin{aligned} \mathcal{C}_{f,g}^{A_1,A_2,\eta}(t,u) = \\ \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2,u_2) ds, \\ b_1, b_2 \neq 0, \end{aligned} \quad (6)$$

where $\eta = 0, 1, i, j, k$, $K_{A_1}^\lambda(s_1, u_1)$, and $K_{A_2}^\mu(s_2, u_2)$ are given by (3) and (4), respectively.

B. The relationship between multiple types of WVD-QCOLCTs

Theorem 1. Let f, g be quaternion functions. Then we have

$$\mathcal{C}_{f,g}^{A_1,A_2,0} = \frac{1}{2} \left(\mathcal{C}_{f,g}^{A_1,A_2,i} + \mathcal{C}_{f,g}^{A_1,A_2,j} + \mathcal{C}_{f,g}^{A_1,A_2,k} - \mathcal{C}_{f,g}^{A_1,A_2,1} \right). \quad (7)$$

Proof. Let $g\left(t - \frac{s}{2}\right) = g_0 + g_1 i + g_2 j + g_3 k$, where $g_0, g_1, g_2, g_3 \in \mathbb{R}$. Therefore, we have

$$\begin{aligned} \mathcal{C}_{f,g}^{A_1,A_2,0} &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times (g_0 + g_1 i + g_2 j + g_3 k) K_{A_2}^\mu(s_2,u_1) ds, \\ \mathcal{C}_{f,g}^{A_1,A_2,1} &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times (g_0 - g_1 i - g_2 j - g_3 k) K_{A_2}^\mu(s_1,u_1) ds, \\ \mathcal{C}_{f,g}^{A_1,A_2,i} &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times (g_0 - g_1 i + g_2 j + g_3 k) K_{A_2}^\mu(s_1,u_1) ds, \end{aligned}$$

$$\begin{aligned} \mathcal{C}_{f,g}^{A_1,A_2,j} &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times (g_0 + g_1 i - g_2 j + g_3 k) K_{A_2}^\mu(s_1,u_1) ds, \\ \mathcal{C}_{f,g}^{A_1,A_2,k} &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times (g_0 + g_1 i + g_2 j - g_3 k) K_{A_2}^\mu(s_1,u_1) ds. \end{aligned}$$

Thus

$$\begin{aligned} &\frac{1}{2} \left(\mathcal{C}_{f,g}^{A_1,A_2,i} + \mathcal{C}_{f,g}^{A_1,A_2,j} + \mathcal{C}_{f,g}^{A_1,A_2,k} - \mathcal{C}_{f,g}^{A_1,A_2,1} \right) \\ &= \frac{1}{2} \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times \begin{pmatrix} g_0 - g_1 i + g_2 j + g_3 k \\ +g_0 + g_1 i - g_2 j + g_3 k \\ +g_0 + g_1 i + g_2 j - g_3 k \\ -g_0 + g_1 i + g_2 j + g_3 k \end{pmatrix} K_{A_2}^\mu(s_1,u_1) ds \\ &= \frac{1}{2} \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2}\right) \\ &\quad \times 2(g_0 + g_1 i + g_2 j + g_3 k) K_{A_2}^\mu(s_1,u_1) ds \\ &= \mathcal{C}_{f,g}^{A_1,A_2,0}. \end{aligned}$$

Theorem 1 shows that the WVD-QCOLCT of any correlation can be linearly expressed by the WVD-QCOLCT of four other distinct correlations.

C. Some properties of WVD-QCOLCT

We derive some important properties of the WVD-QCOLCT:

Theorem 2. (Shift) Let f and g be quaternion functions. Then the following holds:

$$\mathcal{C}_{f(t-m),g(t-m)}^{A_1,A_2,\eta}(t,u) = \mathcal{C}_{f(t),g(t)}^{A_1,A_2,\eta}(t-m,u). \quad (8)$$

Proof.

$$\begin{aligned} &\mathcal{C}_{f(t-m),g(t-m)}^{A_1,A_2,\eta}(t,u) \\ &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left(t + \frac{s}{2} - m\right) g^{\eta*}\left(t - \frac{s}{2} - m\right) \\ &\quad \times K_{A_2}^\mu(s_2,u_2) ds \\ &= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1,u_1) f\left((t-m) + \frac{s}{2}\right) g^{\eta*}\left((t-m) - \frac{s}{2}\right) \\ &\quad \times K_{A_2}^\mu(s_2,u_2) ds \\ &= \mathcal{C}_{f(t),g(t)}^{A_1,A_2,\eta}(t-m,u) \end{aligned}$$

Theorem 3. (Dilation) Let f and g be quaternion functions. Then the following holds:

$$\mathcal{C}_{\frac{1}{m}f\left(\frac{t}{m}\right),\frac{1}{m}g\left(\frac{t}{m}\right)}^{A_1,A_2,\eta}(t,u) = \mathcal{C}_{f(t),g(t)}^{B_1,B_2,\eta}\left(\frac{t}{m},\frac{u}{m}\right), \quad (9)$$

where $A_l = \begin{bmatrix} a_l & b_l \\ c_l & d_l \end{bmatrix} \begin{bmatrix} \tau_l \\ \omega_l \end{bmatrix}$, $B_l = \begin{bmatrix} a_l & \frac{b_l}{m^2} \\ m^2 c_l & d_l \end{bmatrix} \begin{bmatrix} \tau_l \\ m \omega_l \end{bmatrix}$, $l = 1, 2$ and $m \neq 0$.

Proof.

$$\begin{aligned}
& \mathcal{C}_{\frac{1}{m}f(\frac{t}{m}), \frac{1}{m}g(\frac{t}{m})}^{A_1, A_2, \eta}(t, u) \\
&= \frac{1}{m^2} \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) f\left(\frac{t}{m} + \frac{s}{2m}\right) g^{\eta*}\left(\frac{t}{m} - \frac{s}{2m}\right) \\
&\quad \times K_{A_2}^\mu(s_2, u_2) ds \\
&\stackrel{\text{let } y = \frac{s}{m}}{=} \int_{\mathbb{R}^2} K_{A_1}^\lambda(my_1, u_1) f\left(\frac{t}{m} + \frac{y}{2}\right) g^{\eta*}\left(\frac{t}{m} - \frac{y}{2}\right) \\
&\quad \times K_{A_2}^\mu(my_2, u_2) dy \\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \lambda}} \\
&\quad \times e^{\frac{\lambda}{2b_1} [a_1 m^2 y_1^2 - 2my_1(u_1 - \tau_1) - 2u_1(d_1 \tau_1 - b_1 w_1) + d_1(u_1^2 + \tau_1^2)]} \\
&\quad \times f\left(\frac{t}{m} + \frac{y}{2}\right) g^{\eta*}\left(\frac{t}{m} - \frac{y}{2}\right) \frac{1}{\sqrt{2\pi b_2 \mu}} \\
&\quad \times e^{\frac{\mu}{2b_2} [a_2 m^2 y_2^2 - 2my_2(u_2 - \tau_2) - 2u_2(d_2 \tau_2 - b_2 w_2) + d_2(u_2^2 + \tau_2^2)]} dy \\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \lambda}} \\
&\quad \times e^{\frac{\lambda}{2b_1} \left[\frac{a_1 y_1^2}{m^2} - \frac{2y_1}{m^2} \left(\frac{u_1}{m} - \frac{\tau_1}{m} \right) - \frac{2}{m^2} \frac{u_1}{m} \left(d_1 \frac{\tau_1}{m} - \frac{b_1}{m^2} m w_1 \right) + \frac{d_1}{m^2} \left(\left(\frac{u_1}{m} \right)^2 + \left(\frac{\tau_1}{m} \right)^2 \right) \right]} \\
&\quad \times f\left(\frac{t}{m} + \frac{y}{2}\right) g^{\eta*}\left(\frac{t}{m} - \frac{y}{2}\right) \frac{1}{\sqrt{2\pi b_2 \mu}} \\
&\quad \times e^{\frac{\mu}{2b_2} \left[\frac{a_2 y_2^2}{m^2} - \frac{2y_2}{m^2} \left(\frac{u_2}{m} - \frac{\tau_2}{m} \right) - \frac{2}{m^2} \frac{u_2}{m} \left(d_2 \frac{\tau_2}{m} - \frac{b_2}{m^2} m w_2 \right) + \frac{d_2}{m^2} \left(\left(\frac{u_2}{m} \right)^2 + \left(\frac{\tau_2}{m} \right)^2 \right) \right]} dy \\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1 \lambda}} \\
&\quad \times e^{\frac{\lambda}{2} \frac{b_1}{m^2} \left[\frac{a_1 y_1^2}{m^2} - 2y_1 \left(\frac{u_1}{m} - \frac{\tau_1}{m} \right) - 2 \frac{u_1}{m} \left(d_1 \frac{\tau_1}{m} - \frac{b_1}{m^2} m w_1 \right) + d_1 \left(\left(\frac{u_1}{m} \right)^2 + \left(\frac{\tau_1}{m} \right)^2 \right) \right]} \\
&\quad \times f\left(\frac{t}{m} + \frac{y}{2}\right) g^{\eta*}\left(\frac{t}{m} - \frac{y}{2}\right) \frac{1}{\sqrt{2\pi b_2 \mu}} \\
&\quad \times e^{\frac{\mu}{2} \frac{b_2}{m^2} \left[\frac{a_2 y_2^2}{m^2} - 2y_2 \left(\frac{u_2}{m} - \frac{\tau_2}{m} \right) - 2 \frac{u_2}{m} \left(d_2 \frac{\tau_2}{m} - \frac{b_2}{m^2} m w_2 \right) + d_2 \left(\left(\frac{u_2}{m} \right)^2 + \left(\frac{\tau_2}{m} \right)^2 \right) \right]} dy \\
&= \int_{\mathbb{R}^2} K_{B_1}^\lambda\left(y_1, \frac{u_1}{m}\right) f\left(\frac{t}{m} + \frac{y}{2}\right) g^{\eta*}\left(\frac{t}{m} - \frac{y}{2}\right) \\
&\quad \times K_{B_1}^\mu\left(y_2, \frac{u_2}{m}\right) dy \\
&= \mathcal{C}_{f(t), g(t)}^{B_1, B_2, \eta}\left(\frac{t}{m}, \frac{u}{m}\right),
\end{aligned}$$

where $B = \begin{bmatrix} a_l & \frac{b_l}{m^2} \\ m^2 c_l & d_l \end{bmatrix}, l = 1, 2.$

Theorem 4. (Nonlinearity) Let f and g be quaternion func-

tions. Then the following holds:

$$\mathcal{C}_{f+g, f+g}^{A_1, A_2, \eta} = \mathcal{C}_{f, f}^{A_1, A_2, \eta} + \mathcal{C}_{f, g}^{A_1, A_2, \eta} + \mathcal{C}_{g, f}^{A_1, A_2, \eta} + \mathcal{C}_{g, g}^{A_1, A_2, \eta}. \quad (10)$$

Proof.

$$\begin{aligned}
& \mathcal{C}_{f+g, f+g}^{A_1, A_2, \eta}(t, u) \\
&= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) \left[f\left(t + \frac{s}{2}\right) + g\left(t + \frac{s}{2}\right) \right] \\
&\quad \times \left[f^{\eta*}\left(t - \frac{s}{2}\right) + g^{\eta*}\left(t - \frac{s}{2}\right) \right] K_{A_2}^\mu(s_2, u_2) ds \\
&= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) \\
&\quad \times \left[f\left(t + \frac{s}{2}\right) f^{\eta*}\left(t - \frac{s}{2}\right) + f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) \right. \\
&\quad \left. + g\left(t + \frac{s}{2}\right) f^{\eta*}\left(t - \frac{s}{2}\right) + g\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) \right] \\
&\quad \times K_{A_2}^\mu(s_2, u_2) ds \\
&= \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) f\left(t + \frac{s}{2}\right) f^{\eta*}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2, u_2) ds \\
&\quad + \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2, u_2) ds \\
&\quad + \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) g\left(t + \frac{s}{2}\right) f^{\eta*}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2, u_2) ds \\
&\quad + \int_{\mathbb{R}^2} K_{A_1}^\lambda(s_1, u_1) g\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) K_{A_2}^\mu(s_2, u_2) ds \\
&= \mathcal{C}_{f, f}^{A_1, A_2, \eta} + \mathcal{C}_{f, g}^{A_1, A_2, \eta} + \mathcal{C}_{g, f}^{A_1, A_2, \eta} + \mathcal{C}_{g, g}^{A_1, A_2, \eta}.
\end{aligned}$$

Remark 5. It is evident that if we set

$$h_{f, g}^\eta(t, s) = f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right), \quad (\eta \in \{0, 1, i, j, k\}),$$

then the following holds:

$$\mathcal{C}_{f, g}^{A_1, A_2, \eta}(t, u) = \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \left\{ h_{f, g}^\eta(t, s) \right\}(u). \quad (11)$$

We proceed to present the inversion for the WVD-QCOLCT.

Theorem 6. Let f , g , and $\mathcal{C}_{f, g}^{A_1, A_2, \eta}$ be quaternion functions, then the inverse transform of the WVD-QCOLCT is expressed as

$$\begin{aligned}
& f(v) = \\
& \frac{1}{|g^{\eta*}|_{2, Q}^2} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \overline{K_{A_1}^\lambda\left(\frac{v_1 + \varepsilon_1}{2}, u_1\right)} \mathcal{C}_{f, g}^{A_1, A_2, \eta}\left(\frac{v + \varepsilon}{2}, u\right) \\
& \quad \times \overline{K_{A_2}^\mu\left(\frac{v_2 + \varepsilon_2}{2}, u_2\right)} (g^{\eta*}(\varepsilon))^{1*} du d\varepsilon.
\end{aligned} \quad (12)$$

Proof. By equation (11)

$$\mathcal{C}_{f, g}^{A_1, A_2, \eta}(t, u) = \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \left\{ f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) \right\}(u)$$

and by [18],

$$\begin{aligned}
& f\left(t + \frac{s}{2}\right) g^{\eta*}\left(t - \frac{s}{2}\right) = \\
& \int_{\mathbb{R}^2} \overline{K_{A_1}^\lambda(t_1, u_1)} \mathcal{C}_{f, g}^{A_1, A_2, \eta}(t, u) \overline{K_{A_2}^\mu(t_2, u_2)} du.
\end{aligned}$$

By setting $r = t + \frac{s}{2}$, $w = t - \frac{s}{2}$, we obtain $t = \frac{r+w}{2}$ and the following:

$$\begin{aligned} f(r)g^{\eta*}(w) &= \\ \int_{\mathbb{R}^2} \overline{K_{A_1}^\lambda \left(\frac{r_1 + w_1}{2}, u_1 \right)} \mathcal{C}_{f,g}^{A_1, A_2, \eta} \left(\frac{r + w}{2}, u \right) \\ &\times K_{A_2}^\mu \left(\frac{r_2 + w_2}{2}, u_2 \right) du. \end{aligned} \quad (13)$$

After multiplying both sides of (13) on the right by $(g^{\eta*}(w))^{1*}$ and integrating with respect to $d\varepsilon$, we have:

$$\begin{aligned} f(r) \int_{\mathbb{R}^2} |g^{\eta*}(w)|^2 dw &= \\ \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \overline{K_{A_1}^\lambda \left(\frac{r_1 + w_1}{2}, u_1 \right)} \mathcal{C}_{f,g}^{A_1, A_2, \eta} \left(\frac{r + w}{2}, u \right) \\ &\times K_{A_2}^\mu \left(\frac{r_2 + w_2}{2}, u_2 \right) (g^{\eta*}(w))^{1*} dudw. \end{aligned}$$

Consequently

$$\begin{aligned} f(r) &= \frac{1}{|g^{\eta*}|_{2,Q}^2} \\ &\times \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \overline{K_{A_1}^\lambda \left(\frac{r_1 + w_1}{2}, u_1 \right)} \mathcal{C}_{f,g}^{A_1, A_2, \eta} \left(\frac{r + w}{2}, u \right) \\ &\times K_{A_2}^\mu \left(\frac{r_2 + w_2}{2}, u_2 \right) (g^{\eta*}(w))^{1*} dudw. \end{aligned}$$

Theorem 7. (Plancherel's theorem for the WVD-QCOLCT). Let f, g be quaternion functions. Then we have

$$\left\| \mathcal{C}_{f,g}^{A_1, A_2, \eta} \right\|_{2,Q} = |f|_{2,Q}^2 |g^{\eta*}|_{2,Q}^2. \quad (14)$$

Proof. By equality (11) we have

$$\mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, u) = \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{h_{f,g}^\eta(t, s)\}(u)$$

and the Plancherel formula for the QOLCT [14]

$$\left\| \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{f\} \right\|_{2,Q} = |f|_{2,Q}.$$

So

$$\begin{aligned} \left\| \mathcal{C}_{f,g}^{A_1, A_2, \eta} \right\|_{2,Q} &= \left\| \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{h_{f,g}^\eta\} \right\|_{2,Q} = |h_{f,g}^\eta|_{2,Q} \\ &= \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left| f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_Q^2 dt ds \right)^{\frac{1}{2}} \\ &\stackrel{u=t+\frac{s}{2}, v=t-\frac{s}{2}}{=} \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} |f(u)g^{\eta*}(v)|_Q^2 dudv \right)^{\frac{1}{2}} \\ &= \int_{\mathbb{R}^2} |f(u)|^2 du \int_{\mathbb{R}^2} |g^{\eta*}(v)|^2 dv = |f|_{2,Q}^2 |g^{\eta*}|_{2,Q}^2. \end{aligned}$$

D. The uncertainty principle of the WVD-QCOLCT

We give the uncertainty principle for the WVD-QCOLC.

Theorem 8. (Heisenberg QOLCT [18]). Let $f, \frac{\partial}{\partial s_k} f, s_k f$ be quaternion functions, for $k = 1, 2$. Then the following holds:

$$\begin{aligned} |s_k f(s)|_{2,Q}^2 &\left\| \frac{\xi_k}{2\pi b_k} \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{f(s)\}(\xi) \right\|_{2,Q}^2 \\ &\geq \frac{1}{16\pi^2} |f(s)|_{2,Q}^4. \end{aligned} \quad (15)$$

Theorem 9. (Heisenberg WVD-QCOLCT). Let f, g be quaternion functions. We obtain the inequality

$$\begin{aligned} &\left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left| s_k f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_Q^2 ds dt \right) \\ &\times \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_Q^2 d\xi dt \right) \\ &\geq \frac{1}{16\pi^2} |f|_{2,Q}^4 |g^{\eta*}|_{2,Q}^4. \end{aligned} \quad (16)$$

Proof. Let $h_{f,g}^\eta$ be expressed as specified in Remark 5. Therefore, by applying (15), we get

$$\begin{aligned} &\left| s_k h_{f,g}^\eta(., s) \right|_{2,Q}^2 \left\| \frac{\xi_k}{2\pi b_k} \mathcal{O}_{A_1, A_2}^{\lambda, \mu} \{h_{f,g}^\eta(., s)\}(\xi) \right\|_{2,Q}^2 \\ &\geq \frac{1}{16\pi^2} |h_{f,g}^\eta(., s)|_{2,Q}^4. \end{aligned}$$

According to (11), we obtain

$$\begin{aligned} &\left| s_k h_{f,g}^\eta(t, s) \right|_{2,Q}^2 \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_{2,Q}^2 \\ &\geq \frac{1}{16\pi^2} |h_{f,g}^\eta(t, s)|_{2,Q}^4. \end{aligned}$$

Then we have

$$\begin{aligned} &\left| s_k f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_{2,Q}^2 \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_{2,Q}^2 \\ &\geq \frac{1}{16\pi^2} |h_{f,g}^\eta(t, s)|_{2,Q}^4. \end{aligned} \quad (17)$$

Take the square of inequality (17) on both sides, and then integrate dt . We obtain

$$\begin{aligned} &\int_{\mathbb{R}^2} \left(\left(\int_{\mathbb{R}^2} \left| s_k f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_Q^2 ds \right)^{\frac{1}{2}} \right. \\ &\times \left. \left(\int_{\mathbb{R}^2} \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_Q^2 d\xi \right)^{\frac{1}{2}} \right) dt \\ &\geq \frac{1}{4\pi} \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} |h_{f,g}^\eta(t, s)|_Q^2 ds dt. \end{aligned} \quad (18)$$

We apply the Schwartz inequality to the left-hand side of (18) and make use of (14). We get

$$\begin{aligned} &\left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left| s_k f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_Q^2 ds dt \right)^{\frac{1}{2}} \\ &\times \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_Q^2 d\xi dt \right)^{\frac{1}{2}} \\ &\geq \frac{1}{4\pi} |f|_{2,Q}^2 |g^{\eta*}|_{2,Q}^2. \end{aligned} \quad (19)$$

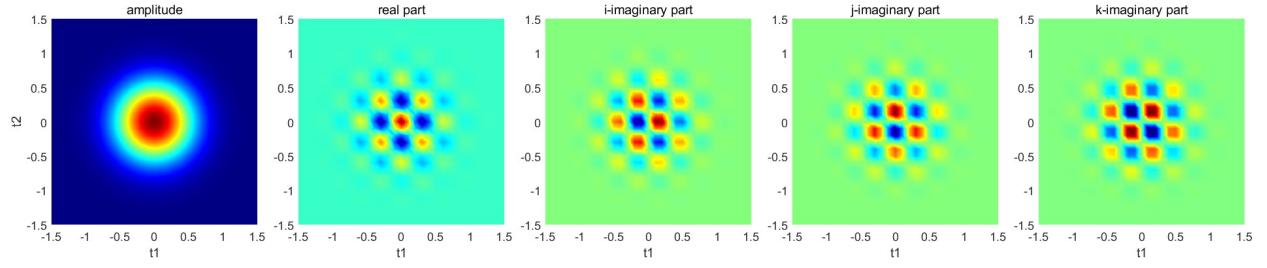


Fig. 1: The original signal of f (21)

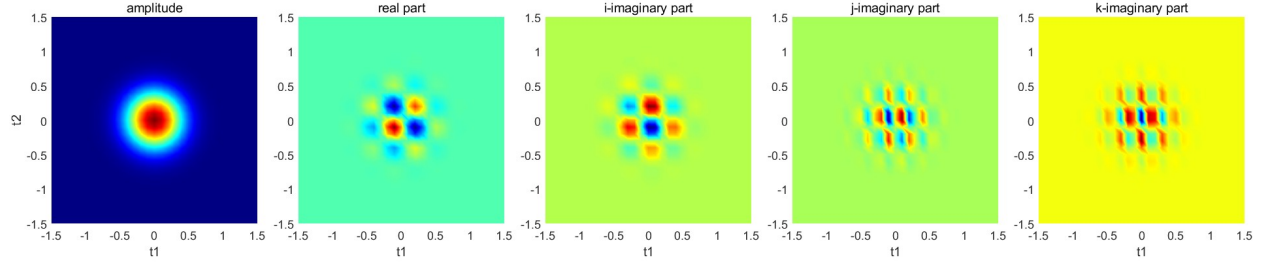


Fig. 2: The 1 correlation $h_{f,f}^1$ of f (21)

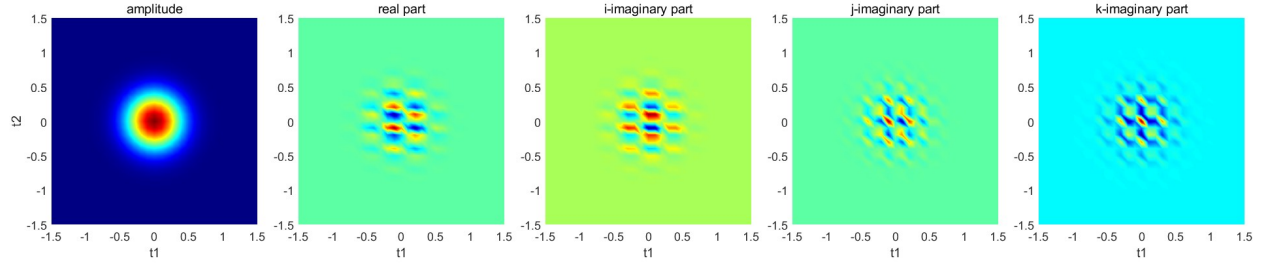


Fig. 3: The i correlation $h_{f,f}^i$ of f (21)

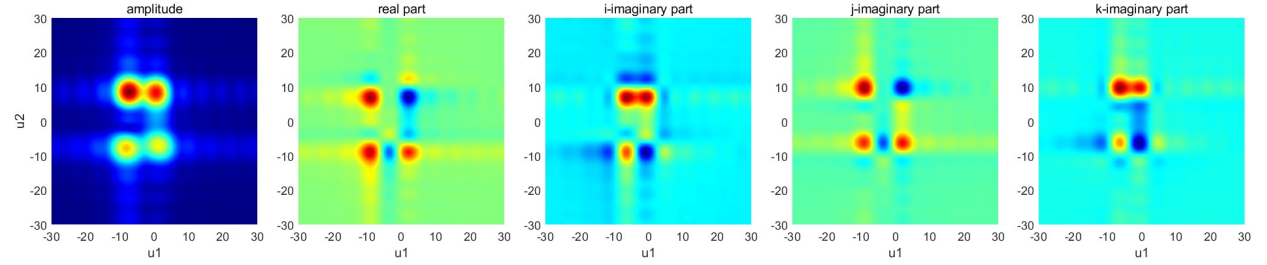


Fig. 4: The WVD-QCOLCT of $C_{f,f}^{A_1, A_2, 1}$ of f (21)

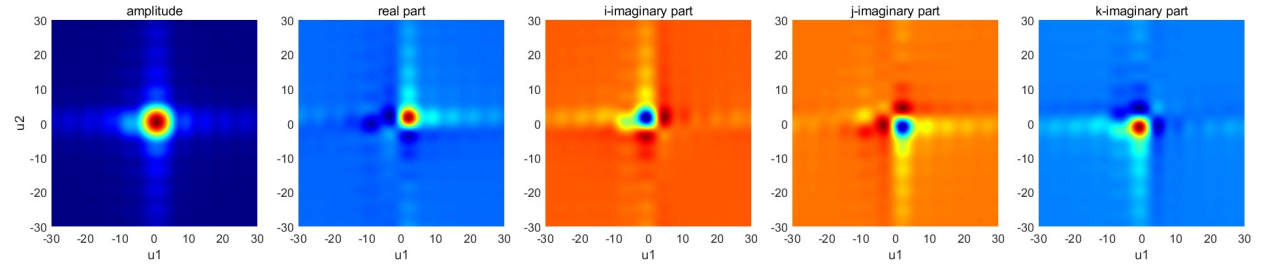


Fig. 5: The WVD-QCOLCT of $C_{f,f}^{A_1, A_2, i}$ of f (21)

Take the square of inequality (19) on both sides to obtain the Heisenberg inequality:

$$\begin{aligned} & \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left| s_k f \left(t + \frac{s}{2} \right) g^{\eta*} \left(t - \frac{s}{2} \right) \right|_Q^2 ds dt \right) \\ & \times \left(\int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \left\| \frac{\xi_k}{2\pi b_k} \mathcal{C}_{f,g}^{A_1, A_2, \eta}(t, \xi) \right\|_Q^2 d\xi dt \right) \\ & \geq \frac{1}{16\pi^2} |f|_{2,Q}^4 |g^{\eta*}|_{2,Q}^4. \end{aligned} \quad (20)$$

IV. SIGNAL DETECTION OF QUATERNION SIGNALS

In this section, we will demonstrate that by selecting the appropriate η correlation, the WVD-QCOLCT can perform signal detection on signal $f(t_1, t_2) \in L^1(\mathbb{R}^2, \mathbb{H})$ (see (21))

$$f(t_1, t_2) = e^{-\pi(t_1^2 + t_2^2)} e^{i\left(\frac{a_g}{2b_g} t_1^2 - \frac{1}{b_g} t_1 \gamma_1 + \frac{d_g}{2b_g} \gamma_1^2\right)} e^{j\left(\frac{a'_g}{2b'_g} t_2^2 - \frac{1}{b'_g} t_2 \gamma_2 + \frac{d'_g}{2b'_g} \gamma_2^2\right)}, \quad (21)$$

where

$$\begin{pmatrix} a_g & b_g \\ d_g & d_g \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} a'_g & b'_g \\ c_g & d'_g \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \\ \gamma_1 = 10, \gamma_2 = 10.$$

For this quaternion signal, we find that it is dispersed in the (t_1, t_2) time domain Fig. 1. Under multiple correlation conditions, the signal distribution is also dispersed in the (t_1, t_2) time domain Fig. 2, Fig. 3.

For the WVD-QCOLCT transform (5), we take parameters as $A_1 = A_2 = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. From Fig. 4, we can see that for signal f , taking its 1 correlation and performing the WVD-QCOLCT, with a fixed time domain (t_1, t_2) , the signal exhibits multiple peaks and valleys in the (u_1, u_2) frequency domain. There are still interference terms in the frequency domain. Therefore, it is difficult to perform signal detection. But when we perform the WVD-QCOLCT on the i correlation of signal f , the energy of the signal is concentrated within the frequency domain (u_1, u_2) over a fixed time domain (t_1, t_2) Fig. 5. Due to the presence of a single energy peak in (u_1, u_2) the frequency domain, we can achieve signal detection in the (u_1, u_2) frequency domain. This example demonstrates that selecting the appropriate η correlation, and performing the WVD-QCOLCT on the signal can achieve signal detection.

V. CONCLUSION

This paper incorporates multiple quaternion correlations into the Wigner-Ville distribution related to the quaternionic offset linear canonical transform (WVD-QOLCT), and puts forward a new type of WVD with multiple quaternion correlations based on the offset linear canonical transform (WVD-QCOLCT). Afterwards, this paper derives the relationship between the WVD-QCOLCT under five quaternion correlations. Some important properties such as time shift, dilation, and Plancherel's theory have been proven. In addition, the uncertainty principle of the WVD-QCOLCT has been established.

Finally, this paper provides an example of signal detection using the WVD-QCOLCT. The results indicate that selecting an appropriate quaternion correlation can enable the signal to accumulate energy in the WVD-QCOLCT domain, thereby completing signal detection.

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